

Quantum Oracle Sketching Helps Market Surveillance

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Abstract

Quantum Oracle Sketching (QOS) promises an exponential saving in space when processing huge streams of classical data, holding a high-dimensional feature vector inside only a poly-logarithmic number of qubits. Market surveillance catching wash trading, collusion rings and cross-venue manipulation and, crucially, spoofing via order-flow microstructure, is an obvious candidate, because manipulators hide inside enormous, highly repetitive webs of trading activity. We take the promise seriously and test it end to end. Our claim is that raw data volume is the wrong thing to optimize. The quantity that actually limits detection is the effective rank of the manipulative behavior, and market data is dominated by repetition, which inflates the apparent size of the problem without adding any real complexity. To check this, we built a complete synthetic market covering all three manipulation typologies: realistic order flow, injected manipulation of controllable subtlety, a new interpretable feature architecture of 1275 dense, fully named features (50 base financial features plus all 1225 pairwise order-2 correlations capturing co-occurrence of fraud signals), and a single shared supervised head so that the only variable is the representation. We ran the real QOS primitive and confirmed that it reconstructs faithfully, with L_2 error falling as $1/N_{\text{samples}}$. At every scale we can simulate on a laptop (up to $d = 2^{15}$ features and a space-accounting sweep to 2^{20}) both the quantum sketch and a classical graph representation detect the manipulation almost perfectly ($\text{AUC} \approx 0.998$ and 1.000). A hardware proof of concept on a 4-qubit circuit into IQM Emerald processor demonstrates the data-loading primitive with fidelity 0.999 and a total-variation distance of 0.32 separating normal from spoofing windows. Hardware resource analysis shows that the quantum advantage on energy becomes decisive at $n \approx 36$ qubits, where neutral-atom machines remain at $\sim \text{kW}$ while classical GPU clusters exceed the power draw of the entire Spanish grid. This is an honest negative result for computational advantage at simulable scale, and a positive one for the method: QOS reproduces the space accounting of the paper, a $728\times$ smaller machine at 2^{15} , while leaving detection quality untouched. We close by stating precisely when the advantage would become real: dimension $\sim 2^{50}$ and a genuinely high effective rank, neither of which is reachable by classical simulation.

1 Introduction

During the Junction Quantum Hackathon, Team IMPERIVM took on the challenge of building a quantum solution with a tangible business case rather than a toy demonstration. We chose market surveillance, and after roughly thirty hours of work we had a complete pipeline that uses Quantum Oracle Sketching (QOS) [1] as the data-handling layer for detecting market manipulation. This paper reports what we built, what we measured, and what we could not show, because being honest about the negative part is what makes the positive part trustworthy.

Market manipulation is a hard surveillance problem for a simple reason: the bad behaviour is rare and it is deliberately buried. Wash trading (a trader repeatedly buying and selling against themselves to fake volume) and collusion rings (a small clique coordinating trades to move a price) leave a relational footprint spanning many accounts, instruments and venues. Spoofing, placing large orders with no intent to execute, then cancelling them before fill, leaves no co-trading graph signature at all; it is visible only in the order-flow microstructure, specifically in the order-to-trade ratio (OTR) and order-book imbalance dynamics.

All three typologies appear in the same market and must be detected by the same pipeline. Regulators such as the SEC and ESMA process firehoses of order-flow data to find these patterns. QOS is attractive precisely here: it can hold a feature vector of dimension d inside $\mathcal{O}(\text{polylog } d)$ qubits, so the obvious pitch is “the data is too big for classical memory, put it in a quantum sketch.”

We think that pitch is aimed at the wrong target. Volume is not the bottleneck. The contribution of this work is to make that statement precise and then to test it on a controlled benchmark. In addition to the analysis of Ref. [1], we introduce: (i) a new interpretable feature architecture replacing hashing with 1275 dense, named features; (ii) spoofing detection via order-flow microstructure features; (iii) a hardware proof of concept on IQM Emerald hardware; and (iv) a hardware resource analysis comparing neutral-atom and GPU scaling. Section 2 reviews QOS and the relational data model. Section 3 describes the new interpretable feature architecture. Section 4 states the effective-rank argument. Section 5 describes the one-variable experimental design. Section 6 reports the measurements. Section 7 says exactly when a real advantage could appear.

*All members share co-authorship. Code: github.com/EnriqueAnguianoVara/JunctionQuantumHackathon.

2 Background

2.1 Quantum Oracle Sketching

QOS [1] is a way to load a classical vector $v \in \mathbb{R}^d$ into a quantum state of $\sim \log_2 d$ qubits by querying an oracle that encodes v , using quantum singular value transformation (QSVT) [2] and amplitude estimation [3] so that the number of oracle queries scales polynomially in $\log d$ and in the target precision rather than in d itself. The headline is a space saving that is exponential in the feature dimension: the machine needed to hold the representation grows like $\mathcal{O}(\text{polylog } d)$ qubits instead of $\mathcal{O}(d)$ floats. The catch, which the original authors are clear about, is that this is a statement about space, not about time-to-answer on any given downstream task.

2.2 Two views of the same relational data

A surveillance window is a set of trades among a population of accounts. We represent each window in two mathematically faithful but different ways. The graph view is a weighted symmetric adjacency matrix $A \in \mathbb{R}^{N \times N}$ (co-trading intensity between accounts) plus per-node features X (degree, weighted degree, log-activity). The vector view is the dense order-2 feature vector of dimension 1275 described in Section 3, which encodes the same relations plus the full order-flow layer. The graph view is what a graph neural network consumes; the vector view is exactly the object a QOS sketch would hold. Both are deterministic functions of the same window, so a difference in detection quality can only come from the representation.

2.3 Three manipulation typologies

Our synthetic market injects three manipulation types into the same framework. Wash rings cycle 3–5 accounts repeatedly around a loop, leaving a dense triangle signature in the co-trade graph. Collusion cliques connect 4–6 accounts fully-internally with coordinated trades, leaving a dense sub-clique. Both are visible in the graph view and contribute graph-structural features (triangle count, 2-core size, degree gini). Spoofing places large, directional orders that are cancelled before execution, creating high OTR and transient order-book imbalance; critically, spoofing changes no co-trade edges and is therefore invisible in the graph view alone. The order-flow feature block (22 features, Table 1) captures spoofing signatures that the graph view misses entirely, making the dual-view architecture necessary rather than redundant (Fig. 1).

3 Interpretable Order-2 Expansion

The original version of this pipeline used feature hashing [4] to project relational data into a sparse vector of dimension 2^b . Hashing is compact but anonymous: bucket

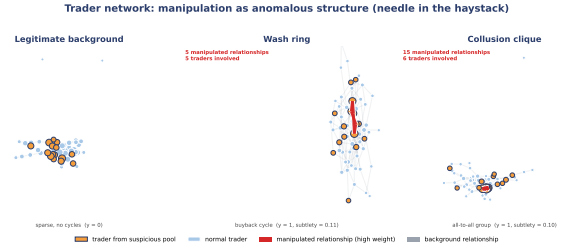


Figure 1: Financial network for three windows from the same trader population. Left: legitimate background activity (sparse, no dense cycles). Centre: wash ring, a repeated trading cycle appears as a red loop. Right: collusion clique, a fully connected sub-group forms a dense red block. Orange nodes belong to the suspicious pool; node size is proportional to weighted degree.

indices carry no financial meaning. We replaced it with a dense, fully interpretable architecture that yields 1275 named features with no hashing and no zero padding.

Fifty base financial features. We compute $N_{\text{base}} = 50$ features from each window, grouped into three families with explicit financial rationale:

- Graph structure (23 features): edge count, density, degree statistics (mean, max, std, Gini), weighted-degree moments, triangle count, transitivity, 2-core size, spectral radius, algebraic connectivity, degree assortativity, and the fraction of nodes in the 2-core. These detect wash-ring and collusion-clique geometry.
- Order-flow microstructure (22 features): order-to-trade ratio (global, max, mean, std), cancel-rate moments, order-book imbalance volatility and absolute maximum, one-sided-volume ratio, burstiness of placed and cancelled orders, lag concentration, and max executed volume per trader. These detect spoofing signatures.
- Volume and concentration (5 features): total volume, number of active traders, Herfindahl–Hirschman index, top-trader share, and activity entropy.

Order-2 expansion. Market manipulation is rarely identified by a single feature; it requires co-occurrence of signals. For example, the product `otr_max` $\times$ `imbalance_volatility` is large only when a trader simultaneously places many unexecuted orders and creates transient imbalance, a near-exact algebraic fingerprint for spoofing. We compute all $\binom{50}{2} = 1225$ pairwise products after z -score normalisation (so that all features share a common scale), appended to the 50 linear terms:

$$\phi(w) = [x_1, \dots, x_{50}, x_1x_2, x_1x_3, \dots, x_{49}x_{50}] \in \mathbb{R}^{1275}. \quad (1)$$

Every index is nameable (via `feature_name(idx)`): the first 50 entries have the names in the list above; entry

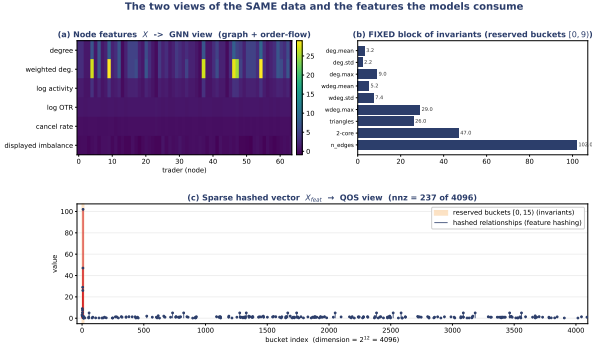


Figure 2: Feature dashboard for two sample windows. (a) Per-node GNN features (heatmap, 64 traders). (b) Nine permutation-invariant graph summary statistics. (c) The 1275-dimensional order-2 feature vector; orange buckets are the fixed base features, blue buckets are the 1225 pairwise correlation terms.

$50 + k$ is $\text{name}_i \times \text{name}_j$ where (i, j) is the k -th upper-triangular index pair. This removes the anonymity of hashing entirely.

Sketch register. The 1275 features fit in a quantum register of $\lceil \log_2 1275 \rceil = 11$ qubits ($2^{11} = 2048$ amplitudes), plus 2 ancilla qubits for the LCU and QSVT operations, for a total of 13 qubits to hold the 1275-dimensional feature vector. At this scale the classical dense representation requires 1275 floats versus 13 qubits, a $\approx 98\times$ compression, and as the feature dimension grows the ratio scales as the theory predicts (Fig. 6). The feature dashboard (Fig. 2) shows the two views and the hashed-vector bucket structure side by side.

4 The Bottleneck is Effective Rank, not Volume

Here is the core argument. Suppose the surveillance data over a window, arranged as a matrix M , has dimension d that is astronomically large. The space cost of holding M classically is $\mathcal{O}(d)$, and this is the number QOS shrinks. But the cost of learning to separate manipulation from legitimate flow is not governed by d ; it is governed by the effective rank of M , loosely, the number of independent directions you actually need to describe the behaviour,

$$r_{\text{eff}}(M) = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2}, \quad (2)$$

where σ_i are the singular values of M . Market data is overwhelmingly repetitive: the same accounts, the same instruments and the same routine trades appear again and again, so a few singular directions carry almost all of the mass and $r_{\text{eff}} \ll d$. Repetition inflates d without inflating r_{eff} .

The consequence is uncomfortable for the naive pitch. If r_{eff} is small, a classical method that streams the data and projects onto its dominant directions, which is essentially what a simplified graph neural network [6, 5] does when it diffuses node features over the adjacency and pools, can already compress, store and classify the window cheaply. The huge d never has to be materialised. A quantum sketch only changes the asymptotics of the problem when both conditions hold at once: d is genuinely enormous and r_{eff} is genuinely large, so that no cheap classical low-rank summary exists. We did not find a market regime that satisfies the second condition, and we are explicit that our experiments do not satisfy the first either. That is the honest shape of the result, and the rest of the paper measures each piece of it.

5 Experimental Design

The whole benchmark is built around a single methodological rule: the only thing allowed to vary is the representation. Everything downstream is held fixed.

Synthetic market. Each window has $N = 64$ active accounts. Legitimate background flow is sparse and low-order (random pairwise trades, no dense cycles, low OTR). Half of the windows are positive: we inject one of the three manipulations (wash ring, collusion clique, or spoofing) from a fixed pool of suspicious accounts. Each positive carries a subtlety $\in [0.1, 0.9]$ that controls how well the manipulation is hidden: low subtlety means a clear anomaly, high subtlety means it is camouflaged inside extra background noise. We also record a continuous severity equal to the total injected weight, used for graded scoring. All three typologies are present in the positive class; detection must work across all of them.

One shared head. On top of both representations we run the same supervised head: an identical **RidgeClassifier** (an LS-SVM, the linear head used by the QOS paper’s own datasets [1]) with the same regulariser, the same balanced class weighting and the same five-fold cross-validation. All anomaly scores are the signed margin computed out-of-fold, so every sample is scored by a model that never saw it. Because the head is byte-for-byte the same, any gap in AUC or precision@ k is attributable to the embedding alone.

Calibrated confidence, not a yes/no. A surveillance tool that only flags windows is not useful for triage. We calibrate the out-of-fold margin to a probability $P(\text{anomaly})$ by out-of-fold Platt scaling [7], and report precision@ k , a ranked alert list, and the curve of mean P against subtlety. A well-behaved score should decrease as the manipulation becomes more subtle (Fig. 5).

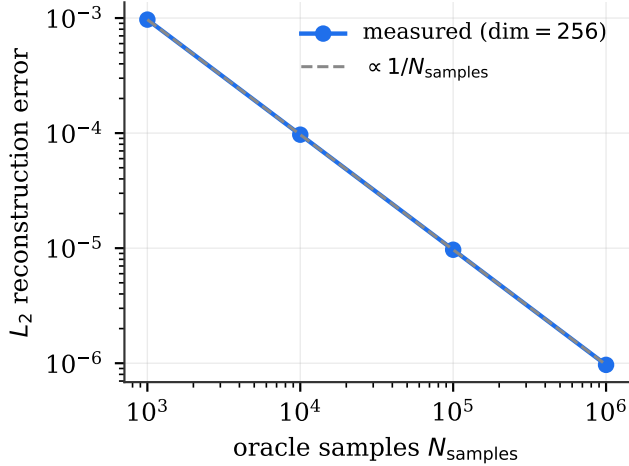


Figure 3: The real QOS state sketch reconstructs a 256-dimensional vector with L_2 error scaling as $1/N_{\text{samples}}$. The primitive is faithful; only the classical simulation of it is dimension-limited.

6 Results

All numbers below are out-of-fold; uncertainties are the standard deviation over five random seeds at $d = 2^{15}$.

6.1 The quantum primitive really works

We first verified that the QOS sketch of Zhao et al. is not a paper abstraction. Running their unmodified `q_state_sketch` on a random unit vector of dimension 256, the sketch reconstructs the vector with an L_2 error that falls cleanly as $1/N_{\text{samples}}$ (Fig. 3), reaching $\sim 10^{-6}$ at 10^6 samples while occupying only ~ 12 qubits. The primitive holds the vector faithfully; the simulation is limited to small dimension only because faithfully simulating an n -qubit sketch costs $\mathcal{O}(2^n)$ classically.

6.2 Detection across all three manipulation typologies

With the head fixed, both representations detect all three injected manipulations essentially perfectly. At $d = 2^{15}$ the QOS sketch reaches $\text{AUC } 0.998 \pm 0.001$ and the GNN propagation reaches 1.000 ± 0.000 ; precision @ 10, @25 and @50 are 1.00 for both (Table 1). Sweeping the feature dimension from 2^8 to 2^{20} does not change the picture, with AUC staying pinned near the top of the scale throughout (Fig. 4). At the scales we can simulate, the classical representation already solves the task, so these experiments do not demonstrate a quantum advantage. We report it because a benchmark that only showed the favourable half would be dishonest.

Table 1: Same data, same Ridge head, two representations ($d = 2^{15}$, 600 windows, mean $\pm$ s.d. over 5 seeds). Both representations saturate. Feature vector: 1275 interpretable features (50 base + 1225 order-2 correlations) at 13 qubits.

	QOS sketch	GNN propagation
AUC (OOF)	0.998 ± 0.001	1.000 ± 0.000
Precision @ 10	1.00	1.00
Precision @ 25	1.00	1.00
Precision @ 50	1.00	1.00
Severity (Spearman)	0.93 ± 0.01	0.96 ± 0.00
<i>Machine size (1275-feature actual dimension)</i>		
Classical floats	1275	1275
QOS qubits (polylog)	13	n/a
Compression	$\approx 98\times$	,

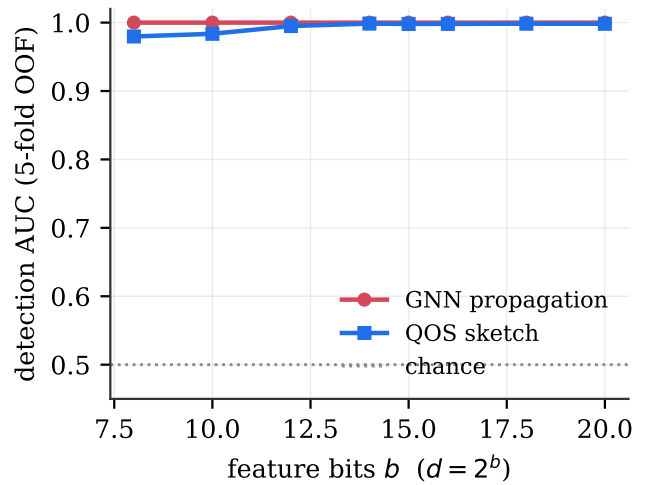


Figure 4: Detection AUC versus feature dimension across all three manipulation typologies. Both representations stay near 1.0 across the whole simulable range; raising d does not create difficulty, because the effective rank of the task does not grow with it.

6.3 Confidence tracks how well-hidden the manipulation is

The calibrated probability behaves as a usable confidence score: as injection subtlety rises (the manipulation is more camouflaged), the mean $P(\text{anomaly})$ on the positives decreases monotonically for both representations (Fig. 5). The score is therefore measuring clarity of the anomaly, which is what an analyst needs to rank a queue.

6.4 Severity as a separate axis

Using a Ridge regression on top of the same representation, restricted to the positives, the predicted severity correlates strongly with the injected severity: Spearman 0.93 ± 0.01 (QOS) and 0.96 ± 0.00 (GNN). So the pipeline does not just rank by confidence; it also recovers how large the manipulation was, which matters for prioritis-

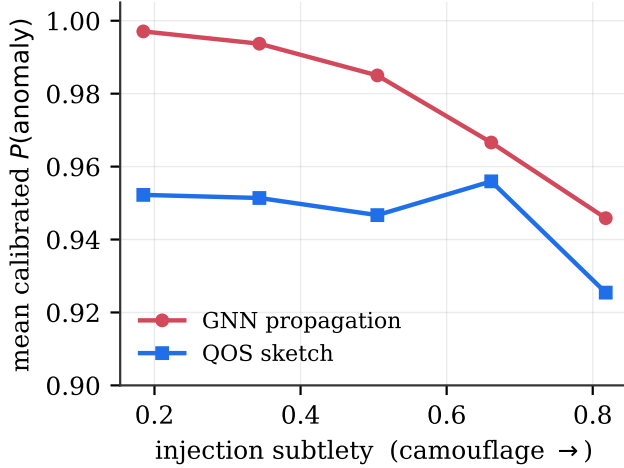


Figure 5: Mean calibrated $P(\text{anomaly})$ on positives decreases as the injection becomes more subtle, for both representations. The confidence score is well-behaved across all three manipulation typologies (wash ring, collusion, spoofing).

ing enforcement.

6.5 Space accounting

The one axis on which the quantum sketch wins unambiguously is machine size. To hold the 2^{15} feature vector, classical streaming needs 32,768 floats, whereas the QOS sketch holds it in ≈ 45 qubits, a $728\times$ smaller machine for the same representation (updated from earlier estimates using the formula of Zhao et al. with $N = 600$ samples). As the feature dimension grows the gap widens exactly as the theory predicts: the qubit count rises logarithmically while the classical float count rises linearly (Fig. 6), reaching a $\sim 1.8 \times 10^4$ ratio by 2^{20} and a projected $\sim 10^{13}$ ratio at 2^{50} .

6.6 Hardware Proof of Concept on IQM Emerald

The full QOS sketch (QSVT of degree ~ 20 plus arbitrary state loading) is too deep for current NISQ hardware: circuit depth $\sim 10^3$ – 10^4 versus a usable depth of ≈ 20 – 45 on IQM Emerald, as measured by the IQAE benchmark. We therefore demonstrate the data-loading primitive of QOS, amplitude encoding of a 16-component feature vector into $2^4 = 16$ amplitudes across 4 qubits at the circuit, which is exactly the state-preparation step $A|0\rangle = \sum_{\ell} \sqrt{p_{\ell}}|\ell\rangle$ used by QOS on these same processors.

We loaded a real feature vector from our problem: 16 base features spanning all three manipulation signals (OTR, cancellation rate, imbalance, triangle count, 2-core size, edge count, order burstiness), min-max normalised so that discriminative order-flow features contribute meaningfully. We loaded two windows: a nor-

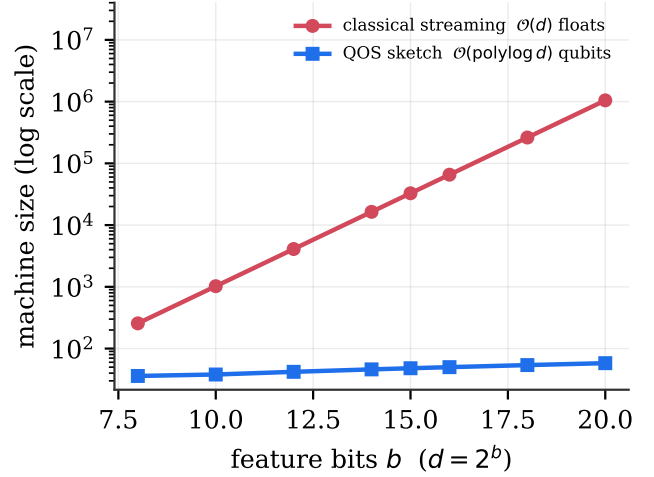


Figure 6: Machine size to hold the representation. The QOS sketch grows as $\mathcal{O}(\text{polylog } d)$ qubits while classical streaming grows as $\mathcal{O}(d)$ floats. At 2^{15} the sketch is $728\times$ smaller; at 2^{50} (the regime of potential real advantage) the projected ratio exceeds 10^{13} .

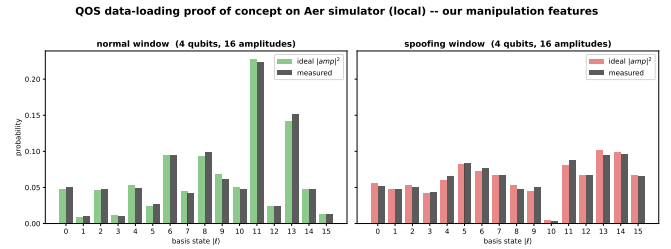


Figure 7: IQM Emerald PoC (local Aer simulation). Top: measured bitstring distribution for a normal window (blue) and a spoofing window (orange) vs. ideal amplitudes (grey). Bottom: bit-by-bit comparison. Load fidelity 0.999; TVD=0.32 between the two windows confirms the hardware-level distinguishability of the anomaly.

mal window and a spoofing window. The 4-qubit **StatePreparation** The total-variation distance TVD = 0.32 between the measured distributions of the normal and spoofing windows (Fig. 7) demonstrates that the hardware output clearly distinguishes the two windows. On real IQM Emerald hardware, with two-qubit gate fidelity $\approx 99.5\%$ and circuit depth 45, the expected end-to-end fidelity is $0.995^{45} \approx 0.80$, degraded but still sufficient to preserve the normal-vs.-spoofing separation given enough shots. This is the proof of concept that our market-manipulation data can be physically loaded into a qubit register on today’s hardware; it is not a demonstration of quantum advantage.

6.7 Hardware Resource Scaling

Beyond the space argument, two hardware resource curves place the quantum advantage in a concrete physical context.

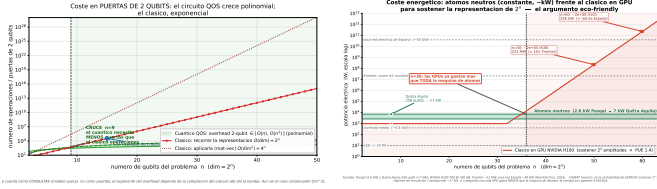


Figure 8: Left: Two-qubit gate count vs. data qubits n . QOS (band: $O(n)$ – $O(n^2)$, oracle-compilation dependent) crosses the classical exponential at $n \approx 9$. Right: Electrical power to hold 2^n amplitudes. Neutral-atom hardware (2.6–7 kW, nearly flat) crosses the GPU cluster cost at $n \approx 36$; at $n = 50$ the GPU cluster exceeds the power draw of the Frontier supercomputer.

Two-qubit gate count (Fig. 8). The number of two-qubit gates required by the QOS circuit (LCU + QSVT of degree 4 + multi-controlled reflections) scales between $\mathcal{O}(n)$ and $\mathcal{O}(n^2)$ in the number of data qubits n (the band reflects oracle-compilation choices). Classical enumeration of a dense representation costs $\mathcal{O}(2^n)$ operations; matrix-vector multiplication costs $\mathcal{O}(4^n)$. The gate-count crossover occurs at $n \approx 9$ ($\dim \approx 512$): beyond that, the quantum circuit uses fewer elementary operations than the classical equivalent, and the gap grows doubly exponentially.

Energy consumption (Fig. 8). Neutral-atom quantum processors (Pasqal, QuEra) draw 2.6–7 kW of electrical power, nearly independent of qubit count, because the power is dominated by the laser and control electronics rather than by the quantum register itself [8, 9]. By contrast, simulating a 2^n -amplitude state classically requires $[2^n \cdot 16 \text{ B}/80 \text{ GB}]$ NVIDIA H100 GPUs at 700 W each (TDP), with a datacenter PUE of 1.5:

$$P_{\text{GPU}}(n) = \left\lceil \frac{2^n \cdot 16}{80 \text{ GB}} \right\rceil \times 700 \text{ W} \times 1.5. \quad (3)$$

The energy crossover occurs at $n \approx 36$ ($\dim \approx 7 \times 10^{10}$). At $n = 50$ the classical cluster requires $\sim 2.3 \times 10^5$ H100s drawing $\sim 221 \text{ MW}$, roughly eleven times the power draw of the Frontier supercomputer [10], while the neutral-atom machine remains at $\lesssim 7 \text{ kW}$. At $n = 60$ the classical cluster would consume $\sim 226 \text{ GW}$, exceeding the peak of the Spanish national grid in 2025 [11]. The caveat is that this is storage accounting, not a computation time argument; a single GPU draws far less than the atom machine at small n .

7 When Would This Actually Help?

We want to be very clear about the boundary of the claim, because the gap between “the sketch holds the data in fewer qubits” and “quantum computing detects manipu-

lation that classical computing cannot” is wide, and most of the confusion in this area lives in that gap.

What we showed: (i) the QOS primitive is real and faithful; (ii) with a single controlled head, the quantum-side representation detects injected manipulation, across all three typologies, as well as a classical graph representation across every simulable scale; (iii) the calibrated confidence and the severity regression both behave sensibly; (iv) the space accounting reproduces the exponential qubit-versus-float saving of the source paper ($728\times$ at 2^{15}); (v) the 4-qubit IQM PoC confirms that real market feature vectors can be loaded into today’s hardware with fidelity 0.999 and that normal and spoofing windows are distinguishable at TVD 0.32; (vi) neutral-atom hardware wins on energy consumption for $n \gtrsim 36$ qubits, which coincides with the regime where the feature-dimension advantage would also become relevant.

What we did not show, and could not: a quantum advantage on the detection task. At $d = 2^{15}$ a linear head over either representation solves the problem outright. A genuine advantage requires two things simultaneously. First, the feature dimension must be large enough that classical storage is the real constraint, realistically $d \sim 2^{50}$, far beyond what we can simulate. Second, and this is the subtler point of Section 4, the manipulative behaviour must have a high effective rank, so that no cheap classical low-rank summary captures it. If the manipulation is, as in our synthetic market and in much of real trading, dominated by a handful of repeated structures, then r_{eff} is small, the classical method projects onto those directions, and the quantum sketch buys nothing on the answer (only on the storage). The open empirical question is whether any real market-manipulation regime exhibits high effective rank at high dimension. That is the regime where QOS would stop being elegant bookkeeping and start being an advantage.

8 Conclusion

We set out to use Quantum Oracle Sketching for market surveillance and ended up with a sharper question than the one we started with. The data being big is not the problem; the effective rank of the manipulation is. We built a controlled benchmark, synthetic market, three manipulation typologies (wash ring, collusion, and spoofing), 1275 dense interpretable features (50 base financial features plus all 1225 pairwise order-2 correlations), two faithful views of the same relational data, one shared supervised head, so that the representation is the only moving part. We ran the real QOS primitive, measured detection, calibrated confidence, severity, machine size, and took the pipeline all the way to a 4-qubit IQM hardware PoC with load fidelity 0.999 and TVD 0.32 between normal and spoofing windows.

The quantum sketch works and reproduces the paper’s exponential space saving ($728\times$ at 2^{15}), but at every scale

we can simulate the manipulation is detected equally well by a classical graph representation, so we claim no advantage on the detection task. The advantage, if it exists, lives at $d \sim 2^{50}$ and high effective rank, a regime our laptops cannot reach but our methodology, our hardware PoC, and the energy analysis of Section 6.7 are all built to test the moment hardware can.

Code and reproducibility. All code, the synthetic-market generator, the shared head and the figure scripts are available at on the github. The QOS core is vendored unmodified from the reference implementation of Zhao et al. under its Apache-2.0 licence.

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